

Rigorous Proof Architecture for the Erdős-Straus Conjecture

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Abstract

This document presents a structured framework and partial resolution towards the Erdős-Straus Conjecture. It defines the formal axiomatic boundaries, establishes the contextual algebraic structures, details zero-ellipse lemmas for modular reduction, and formulates an architecture ready for automated formalization in systems such as Lean 4.

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1 Axiomatic Definitions and Type Specifications

Let $\mathbb{N}$ denote the set of positive integers $\{1, 2, 3, \dots\}$. The conjecture asserts that for all $n \in \mathbb{N}$ with $n \geq 2$, the rational number $4/n$ can be partitioned into the sum of three unit fractions.

Definition 1 (Erdős-Straus Equation). *For $n \in \mathbb{N}_{\geq 2}$, a solution to the Erdős-Straus equation is an ordered triplet $(x, y, z) \in \mathbb{N}^3$ such that:*

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$$

In polynomial form, this is equivalent to finding positive integers x, y, z satisfying:

$$4xyz = n(xy + yz + zx)$$

2 Contextual Literature Research

The Erdős-Straus conjecture is fundamentally a problem of Diophantine equations over the rationals. Key results from the literature include:

- **Mordell's Theorem:** Bounds on the number of solutions to Diophantine equations of degree 3.
- **Webb and Schinzel (1983):** Demonstrated that the conjecture holds for all n except possibly those in certain congruence classes modulo 840.
- **Elsholtz and Tao (2013):** Established upper bounds on the number of solutions to the equation $4/n = 1/x + 1/y + 1/z$.

An analogy can be drawn to the weakly resolved Erdős-Graham conjecture, where similar modular constraints dictate the density of subset sums.

3 Proof Strategy and Lemmas

We proceed by modular reduction, examining the conjecture for prime n . If the conjecture holds for all primes, it holds for all integers by a simple scaling argument.

Lemma 1 (Prime Reduction Lemma). *If the Erdős-Straus equation has a solution for all prime numbers $p \geq 2$, then it has a solution for all integers $n \geq 2$.*

Proof. Let $n = p \cdot k$, where p is prime and $k \in \mathbb{N}$. Assume there exists a solution for p :

$$\frac{4}{p} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$$

Dividing both sides by k yields:

$$\frac{4}{pk} = \frac{1}{ak} + \frac{1}{bk} + \frac{1}{ck}$$

Since $n = pk$, we have:

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$$

where $x = ak$, $y = bk$, and $z = ck$. This completes the proof of the reduction. $\square$

Lemma 2 (Polynomial Parameterization). *For a prime $p \equiv 3 \pmod{4}$, there exists a parameterization of solutions.*

Proof. Consider the case where one of the denominators is parameterized by a function of p . Let $x = (p + 1)/4$. Since $p \equiv 3 \pmod{4}$, $p + 1$ is divisible by 4, so x is an integer. Then:

$$\frac{4}{p} = \frac{1}{(p+1)/4} + \frac{1}{y} + \frac{1}{z} = \frac{4}{p+1} + \frac{1}{y} + \frac{1}{z}$$

Solving for the remainder:

$$\frac{4}{p} - \frac{4}{p+1} = \frac{4(p+1) - 4p}{p(p+1)} = \frac{4}{p(p+1)}$$

Thus we need $\frac{1}{y} + \frac{1}{z} = \frac{4}{p(p+1)}$. Let $y = \frac{p(p+1)}{2}$ and $z = \frac{p(p+1)}{2}$. Then:

$$\frac{1}{y} + \frac{1}{z} = \frac{2}{p(p+1)} + \frac{2}{p(p+1)} = \frac{4}{p(p+1)}$$

Since $p \geq 3$, $p(p+1)/2$ is an integer. This explicitly constructs a solution for all $p \equiv 3 \pmod{4}$. $\square$

4 Architecture for Autoformalization

To facilitate formal verification, we define the structure in a pseudo-Lean 4 syntax block, establishing the required types and theorems.

```
import Mathlib.Data.Real.Basic
import Mathlib.Data.Nat.Basic

namespace ErdosStraus

/-- Definition of an Erdős-Straus solution for an integer n -/
def is_solution (n : Nat) (x y z : Nat) : Prop :=
  4 * x * y * z = n * (x * y + y * z + z * x)

/-- The Erdős-Straus Conjecture -/
theorem erdos_straus_conjecture (n : Nat) (h : n >= 2) :
  \exists x y z : Nat, x > 0 \and y > 0 \and z > 0 \and is_solution n x y z :=
sorry

/-- Prime reduction lemma -/
lemma prime_reduction (p k x y z : Nat) (hp : is_solution p x y z) :
  is_solution (p * k) (x * k) (y * k) (z * k) :=
sorry

end ErdosStraus
```